

Session :- 5

1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Ans:

Task	Git Command
Initialize a repository	git init
Stage all files	git add .
Commit with a message	git commit -m "your message"
Push to remote repository	git push origin main
Pull latest changes	git pull origin main

2. What is the difference between git fetch and git pull?

Answer in 2 lines.

Ans: git fetch downloads the latest changes from the remote repository but does not merge them into the current branch.

git pull downloads the latest changes and automatically merges them into the current branch.

3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans: I can't create a GitHub repository on your behalf, access your local system, or share a repo link from your account.

You can do it with these steps:

1. Create a new repository on GitHub (for example, my-first-repo).

2. Create a small text file locally:

```
echo "Hello GitHub" > sample.txt
```

3. Initialize Git and push the file:

```
git init
```

```
git add sample.txt
```

```
git commit -m "Add sample text file"
```

```
git branch -M main
```

```
git remote add origin https://github.com/<your-username>/my-first-repo.git
```

```
git push -u origin main
```

After pushing, your repository link will be:

```
https://github.com/<your-username>/my-first-repo
```

Replace <your-username> with your GitHub username.

4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Ans: Fork is a copy of someone else's repository created under your own GitHub account so you can make changes independently.

Pull Request is a request to the original repository owner to review and merge your changes into their project.

5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans:

Commands used:

Create and switch to a new branch

git checkout -b feature-test

Make changes and stage the file

git add .

Create one commit on the branch

git commit -m "Add feature-test changes"

Switch back to main branch

git checkout main

Merge the feature branch into main

git merge feature-test